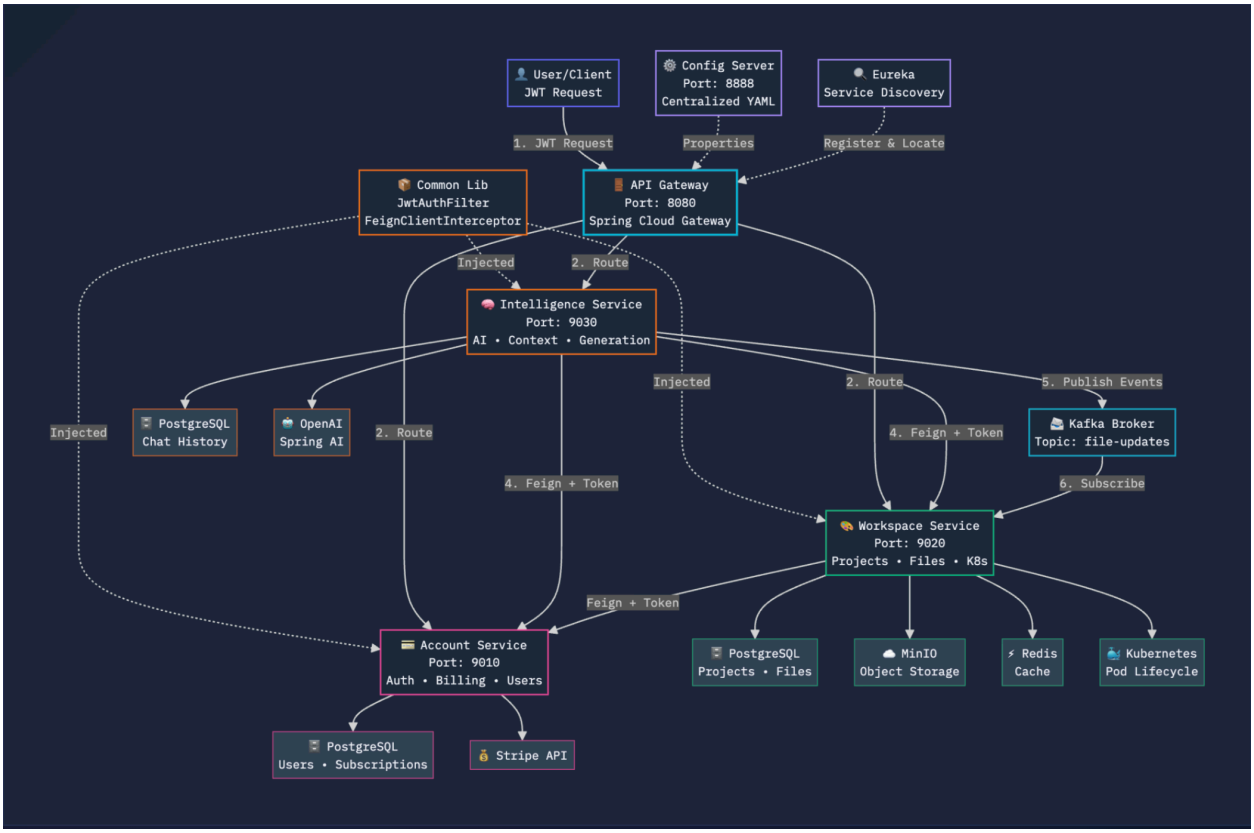
# Distributed Lovable Architecture

## Microservices

| Microservice | Primary Responsibility | Database / Storage | Key Integrations | Cross-Service Calls (Feign) |
|---|---|---|---|---|
| **Account Service** | User Identity, Authentication, and Stripe Billing. | PostgreSQL (Users, Plans, Subscriptions) | Stripe API | *None* |
| **Workspace Service** | Project management, File Tree, Kubernetes Deployments. | PostgreSQL (Projects, Members), **MinIO** (Files) | Kubernetes API, Redis | Calls `AccountService` (Check limits, users) |
| **Intelligence Service** | AI Chat, Context gathering, Code Generation. | PostgreSQL (Chat Sessions, Messages, Events) | OpenAI (Spring AI) | Calls `WorkspaceService` (Read files), `AccountService` (Check AI limits) |




## Account Service

| Category | Components & Details |
|---|---|
| **Dependencies** | `spring-boot-starter-web`, `spring-data-jpa`, `postgresql`, `stripe-java`, `mapstruct`, `spring-cloud-starter-netflix-eureka-client` |
| **Key Entities** | `User`, `Plan` (Free, Pro), `Subscription` |
| **Security Setup** | `AccountSecurityConfig`: Exposes `/auth/**` and `/webhooks/stripe` publicly. Secures everything else. |
| **Internal API** | `InternalAccountController`: Exposes `/internal/v1/...` for other microservices to fetch user DTOs and verify billing plans. |

## Workspace Service

| Category | Components & Details |
|---|---|
| **Dependencies** | `spring-boot-starter-web`, `spring-data-jpa`, `minio`, `kubernetes-client` (Fabric8), `spring-data-redis`, `spring-cloud-starter-openfeign` |
| **Key Entities** | `Project`, `ProjectMember`, `ProjectFile`, `Preview` (K8s pod mapping) |

| | |
|---|---|
| **Cloud Configs** | `StorageConfig` (MinIO connection), `KubernetesConfig`, `RedisConfig` |
| **Security Logic** | `SecurityExpressions`: Contains custom `@PreAuthorize("@security.canEditProject(#id)")` logic to verify ownership via `ProjectMemberRepository`. |
| **Feign Clients** | `AccountClient`: Used during Project Creation to verify if the user has hit their `maxProjects` limit. |

## Intelligence Service

| Category | Components & Details |
|---|---|
| **Dependencies** | `spring-ai-starter-model-openai`, `spring-data-jpa`, `spring-cloud-starter-openfeign` |
| **Key Entities** | `ChatSession`, `ChatMessage`, `ChatEvent` , `UsageLog` (Tracks LLM thoughts, file edits, tool uses) |
| **AI Components** | `AiGenerationServiceImpl` (Prompt execution), `LLMResponseParser` (Regex to extract `<file>` and `<tool>` tags). |
| **Advisors & Tools** | `FileTreeContextAdvisor` (Injects current file tree into prompt), `CodeGenerationHelperTools` (Allows LLM to read specific files). |
| **Feign Clients** | `WorkspaceClient` (To fetch file trees and file content for context), `AccountClient` (To verify AI daily token limits). |

## Common Lib

| Component in common-lib | What it does | Why it's needed |
|---|---|---|
| `JwtAuthFilter` | Extracts the Bearer token and populates the `SecurityContext`. | Every service authenticates requests locally, completely independent of the Gateway. |
| `FeignClientInterceptor` | Automatically grabs the user's JWT from the context and attaches it to outbound Feign requests. | Allows the Intelligence service to call the Workspace service seamlessly as the logged-in user. |
| **Shared DTOs** | `UserDto`, `PlanDto`, `FileTreeDto`, etc. | Prevents JPA `@Entity` classes from leaking across microservice boundaries. |